Problem 11.28

Amin puts 100 into a fund that pays an effective annual rate of discount of 20% for the first two years and a force of interest of rate

$$\delta_t = \frac{2t}{t^2+8}, \qquad 2 \le t \le 4,$$

for the next two years. At the end of four years, the amount in Amin's account is the same as what it would have been if he had put 100 into an account paying interest at the nominal rate of i per annum compounded quarterly for four years. Calculate i.

Solution

During the first two years, the effective annual discount rate is 20%, so the accumulation value is

$$A(2) = 100(1-0.20)^{-2} = 156.25.$$

For the next two years, the accumulation factor under the force of interest is

$$\begin{aligned} A(2,4) &= \exp\left(\int_2^4 \frac{2t}{t^2+8}\,dt\right) \\ &= \exp\left(\ln(t^2+8)\Big|_2^4\right) \\ &= \exp(\ln 24 - \ln 12) = 2. \end{aligned}$$

Thus, the accumulated value of Amin's fund at time 4 is

$$A(4) = 156.25(2) = 312.5.$$

The equivalent account earns interest for four years, or 16 quarterly periods. Equating the two accumulated values,

$$\begin{aligned} 100\left(1+\frac{i}{4}\right)^{16} &= 312.5, \\ i &= 4\left((3.125)^{1/16} - 1\right) \\ &\approx 0.295246748. \end{aligned}$$

Therefore, the nominal annual interest rate compounded quarterly is

$$\boxed{i \approx 0.29525 = 29.5247\%}.$$